Learning with Few Features and Samples

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***Abstract*—Motivated by sensors gathering data for classification purposes, with stringent costs associated to the collection of samples and addition of features, we consider the problem of determining the optimal number of features to be transmitted through the network. Under a bivariate model, we express how the classifier's expected error probability depends on the sample size and on one of the two features' discriminatory power. Then, we use the characterized error probability to determine a threshold on the sample size such that if the sample size is larger than the threshold the presence of an additional feature is in fact beneficial.**

***Index Terms*—Statistical learning, samples, features, SVM.**

I. Introduction

SENSING the environment is a major challenge and a fundamental cornerstone within autonomous systems, vehicular networks, weather forecasting, military networks, and many other modern initiatives. Each sensor collects features from the environment, such as temperature, light, and pressure. After sensors are deployed, they degrade over time. Some sensors will have their batteries depleted, and others will fail. Failures, in turn, will cause a decrease in the number of collected features, where these features may represent different modalities at the same geographical position, or have the same modality but be collected from different geographic locations.

Features and samples have costs associated with them [1]–[6]. Collecting additional features may require the procurement of new sensors, and retrieving more samples per time unit may come at the cost of increased bandwidth and faster depletion of limited batteries. Given such costs, which preclude the collection of a large number of features or the transmission of data at high sampling rates, we pose the following questions:

- given a small sample size, when are fewer features preferred for training a classifier, i.e., when is less better?
- how to compensate for the lack of features using samples?

Motivated by the above questions, in this letter we begin our investigation focusing on the special case of a binary classifier.

Methodology. To illustrate the relationship between sample size, number of features, and asymptotic error probability of the trained classifier, we consider a simple workload with two features. The conditional distribution of each sample given its class is assumed to be bivariate Gaussian (e.g., as in [7], [8]), with a correlation coefficient ρ; one of the features has variance one and the other one has variance δ^2, noting that for $\rho = 0$ the *discriminatory power* of the second feature decreases as δ grows. Although admittedly simple, this model already serves our purposes, namely, to show that *(a)* depending on the discriminatory power of the second feature, it may be worth ignoring it, and *(b)* additional samples can compensate for the lack of a feature.

Contributions. Among our contributions, we express error probability as a function of sample size, number of features, and δ, initially assuming $\rho = 0$. Our analysis reveals the existence of a sample size threshold $n^\star$ such that one should train a classifier using a single feature when the sample size is smaller than $n^\star$, and both features otherwise. The first regime, with very small sample sizes, corresponds to overfitting, wherein fewer features provide greater generalization power and smaller classification error [9], [10]. As sample size grows, the system switches to the second regime, wherein having additional features is helpful, and there is a trade-off between samples and features, i.e., samples can compensate for features. When the number of samples is further increased, however, both samples and features are required to achieve the corresponding accuracy levels. Then, we discuss how $\rho \neq 0$ impacts our results.[1]

II. Problem Formulation

The classification process involves two stages: training the classifier, with labeled samples, and then inferring the classes of unlabeled samples. Our two main attributes of interest are dataset cardinality, denoted by n, which corresponds to the number of samples in the training set, and dataset dimensionality, denoted by d, which corresponds to the number of features in each sample. One of our goals is to assess how n and d simultaneously affect the classifier performance, as measured by its expected error probability.

Background. We consider binary classification with training set $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$, $\boldsymbol{x}_i \in \mathbb{R}^d$, $y_i \in \{-1, +1\}$. A classifier $h : \mathbb{R}^d \to \{-1, +1\}$ has loss $L(h) = \mathbb{P}(Y \neq h(X))$. We focus on linear classifiers and, except otherwise noted, compare one- and two-feature settings.

A dictionary H (also known as hypotheses set) is a set of classifiers (e.g., linear classifiers). The learning problem consists of choosing a classifier from H to be the "solution" for our problem. It is well known that the Bayes classifier, which classifies each input according to $\text{argmax}_y P(Y = y|X = x)$, is globally optimal [12], [13]; the Bayes error is the minimal error probability. We denote by $h^{(B)}$ and $L(h^{(B)})$ the Bayes classifier and its error probability.

Bivariate Gaussian. To assess the generalization power of the considered classifiers, we sample $\mathcal{D}$ from a known probabilistic model, e.g., as in [14], [15]. In particular, we consider a 2-dimensional classification problem, wherein samples from

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[1] A replication package in Python, together with additional details, are available at https://github.com/paulorenatoaz/slacgs [11].

each class are drawn from a bivariate Gaussian distribution. The conditional distribution of feature vector $X = (X_1, X_2)$ given Y is

$$(X_1, X_2) \sim \begin{cases} \mathcal{N}((+1,+1),\Sigma), & \text{if } Y = +1, \\ \mathcal{N}((-1,-1),\Sigma), & \text{if } Y = -1, \end{cases} \tag{1}$$

where $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ denotes a Gaussian distribution with mean vector $\boldsymbol{\mu} \in \mathbb{R}^2$ and covariance matrix Σ,

$$\Sigma = \begin{pmatrix} 1 & \rho\delta \\ \rho\delta & \delta^2 \end{pmatrix}, \tag{2}$$

$\delta \geq 1$ and $|\rho| \leq 1$. Note that δ is the conditional standard deviation of feature X_2, given Y. If $\delta > 1$, feature X_2 has a smaller discriminating power than feature X_1: the larger the value of δ, the less informative is feature X_2. We consider balanced datasets, with $n/2$ samples in each class.

SVM classifier. Our dictionary H is formed by all linear classifiers, i.e., straight lines. A naive approach for learning consists in searching for the line that minimizes the empirical error rate, e.g., through grid search. However, the problem may admit multiple solutions, requiring a principled strategy to break ties [16]. Therefore, we consider linear support vector machines (SVMs), which have a number of desired properties, including (i) a principled strategy to determine the best discriminator, by maximizing the distance, known as the *margin*, between the separating line and the two classes of points to be separated [16]; (ii) polynomial computational cost [17], [18]; more broadly, polynomial approximations (e.g., Taylor/Chebyshev expansions) are a standard route to reduce computational cost in signal processing pipelines [19] and (iii) convergence to an optimal solution, under mild conditions [20]–[22]. We have also experimented with linear discriminant analysis (LDA), obtaining similar results, and opted to report SVM results, since the principles of SVM require less restrictive assumptions [11].

Linear SVM relies on a single parameter, the misclassification cost, appearing as a regularization term in the Lagrange formulation, and denoted by γ [23]. Throughout this work, we let $\gamma = 1$, its default value; we experimented with other values, with results remaining within $\pm 0.5\%$ error.

We denote by $E_d(n; \delta)$ the expected SVM error with n samples and d features, where the conditional standard deviation of X_2 is δ. We analyze the extreme cases $n = \infty$ and $n = 2$ in Section III, and intermediate values in Section IV.

III. Analytical Model

Next, we consider the Bayes error corresponding to an infinite sample size ($n = \infty$). We first derive the theoretical error rate $L_d(\cdot)$ of linear classifiers using d features, and then compare these rates with the corresponding SVM errors.

We start with the case where only feature X_1 is available. We have a 1-dimensional classification problem, wherein we search for the best cut point c. The error probability, under the setting of Section II, is $L_1(c) = (1 + \Phi(c-1) - \Phi(c+1))/2$ where $\Phi(\cdot)$ is the CDF of a Gaussian with mean zero and variance one. The optimal separator is $\tilde{c} = 0$, and $L_1(0) = L_1(\tilde{h}_1) = 1 - \Phi(1) = 0.1586553$.

TABLE I
SVM ERROR PROBABILITIES (MULTIPLIED BY 100) UNDER $\rho = 0$ AS A FUNCTION OF THE SAMPLE SIZE n. THE FIRST BLOCK REPORTS EMPIRICAL ERROR RATES. THE SECOND BLOCK SUMMARIZES A LEAST-SQUARES FIT OF THE FORM $E(n) = \alpha + \beta/n$.

n	1 feature	$E_1(n)$, δ=1	$E_2(n;\delta)$, 2 features δ=2	δ=3	δ=4	δ=5	δ=6	δ=7
2	20.50	14.99	23.29	26.72	28.20	28.93	29.36	29.63
4	18.83	13.91	21.73	24.47	25.62	26.20	26.52	26.73
8	17.63	11.98	18.95	21.02	21.82	22.21	22.42	22.54
16	16.87	10.24	16.28	17.92	18.55	18.86	19.01	19.11
32	16.39	9.18	14.74	16.22	16.78	17.05	17.20	17.29
64	16.15	8.57	13.96	15.39	15.94	16.20	16.34	16.43
129	16.01	8.24	13.57	14.99	15.53	15.79	15.93	16.02
256	15.94	8.06	13.38	14.79	15.33	15.59	15.73	15.82
512	15.90	7.96	13.28	14.69	15.23	15.49	15.63	15.72
1024	15.88	7.91	13.23	14.64	15.18	15.44	15.58	15.67
$n^\star$			12	28	46	70	98	130
Linear regression of error probability: $E(n) = \alpha + \beta/n$								
α	0.159	0.080	0.132	0.146	0.151	0.154	0.155	0.156
β	0.142	0.333	0.468	0.522	0.542	0.552	0.556	0.558
$n^\star$			12	30	52	78	106	136

Next, we consider the case where both features are available. Given a classifier characterized by the line $x_2 = a + bx_1$, its theoretical error rate under the setting of Section II is

$$L_2(a, b) = \frac{1}{2} \sum_{\kappa \in \{+1,-1\}} \left(1 - \Phi\left(\frac{|a + \kappa b - 1|}{\sqrt{S}}\right)\right), \tag{3}$$

where $S = (\rho\delta - b)^2 + \delta^2(1 - \rho^2)$. The optimal solution is given by a straight line through the origin, $\tilde{a} = 0$, with a slope of $\tilde{b} = \delta(\delta - \rho)/(\delta\rho - 1)$. As the classifier $h^{(B)}$ belongs to the dictionary formed by straight lines in $\mathbb{R}^2$, $L(h^{(B)})$ is given by

$$L_2\left(0, \frac{\delta(\delta-\rho)}{\delta\rho - 1}\right) = 1 - \Phi\left(\frac{1}{\delta}\sqrt{\frac{\delta^2 - 2\rho\delta + 1}{1 - \rho^2}}\right). \tag{4}$$

Interestingly, taking the partial derivative of the above expression with respect to ρ and setting it to zero we conclude that the upper bound of $L(h^{(B)})$ is achieved at $\rho = 1/\delta$,

$$0 \leq L(h^{(B)}; \rho) \leq L(h^{(B)}; 1/\delta) = 1 - \Phi(1). \tag{5}$$

The lower bound on the error is achieved for $\rho = 1$ and $\rho = -1$, corresponding to scenarios where X_2 is a deterministic function of X_1. As ρ increases from -1 to $1/\delta$, the error increases; as ρ approaches ± 1, the Bayes error decreases to zero (see also Section IV-C).

In the analysis above, we considered the case $n = \infty$. The opposite extreme, $n = 2$, with a dataset comprised of two samples, one in each class, is also amenable to analytical treatment. For the one-feature case, the expected error is $100 \times \int_{-\infty}^{+\infty} L_1(c)\, \pi^{-1/2} e^{-c^2}\, dc \approx 20.70336$, to be compared against 20.50 in the top-left entry of Table I. For the two-feature case, the error is given by a more complex double integral, where the decision boundary is the perpendicular bisector of the two observed points; numerically, it matches the top row of Table I up to three decimal places. Overall, the top part of Table I suggests that when the sample size is very small (e.g., two observations) and variability is high ($\delta \geq 2$), using fewer features may be advantageous, as discussed below.

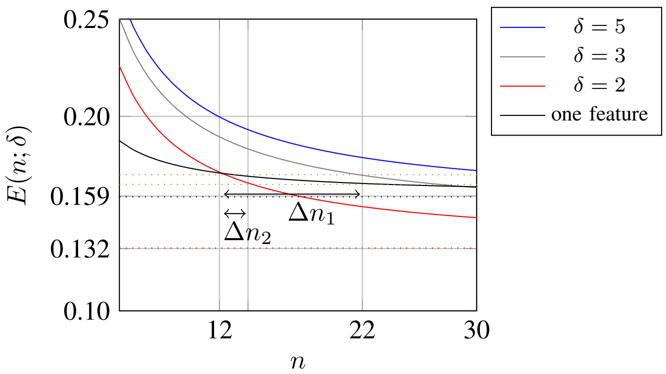

Fig. 1. Fitted error curves $E(n) = \alpha + \beta/n$ for one-feature and two-feature classifiers, for several values of the scale ratio δ. Dashed horizontal lines denote the corresponding asymptotic error levels.

IV. Analyzing the Role of Samples and Features

A. When is it better to use fewer features?

When the sample size is small, using fewer features simplifies training and avoids overfitting. To illustrate this, we compare two scenarios, one with both features and the other with a single feature available, indicating how error probability decays as a function of sample size, for $\rho = 0$.

Table I reports error probabilities (in percent) for SVM classifiers with one feature ($E_1(n)$) and with two features ($E_2(n;\delta)$), across increasing sample sizes n and a range of scale ratios δ. The shaded cells identify the regimes in which adding the second feature strictly reduces the error relative to the one-feature baseline at the same n. To quantify the sample-size dependence compactly, the bottom panel fits each column to $E(n) = \alpha + \beta/n$ and reports a crossover point $n^\star$ for each δ, indicating when (according to the fitted model) the two-feature classifier begins to outperform the one-feature classifier.

More on methodology. Each loss in Table I is obtained from 32,000 datasets sampled from the same distribution. Confidence interval lengths of the error probability estimates were negligible, on the order of 10^{-5}. Recall that each dataset $\mathcal{D}$ contains $n/2$ samples from each class, and that for each of those datasets SVM produces a corresponding separating line (in the case of two features) or cut point (in the case of a single feature). The corresponding error probabilities are computed using the results reported in Sec. III and each set of m error probabilities is averaged to produce the numbers in the table.

Fewer features can be better. Table I shows that $E_1(2) < E_2(2;\delta)$ and, for $\delta \geq 2$, there is a point $n^\star(\delta)$ at which $E_1(n^\star(\delta)) = E_2(n^\star(\delta);\delta)$. Table I indicates that $n^\star(\delta)$ is an increasing function of δ, i.e., the less informative the second feature is, the larger the sample size is needed to justify the use of such an additional feature. Indeed, when n is small, there is overfitting, i.e., the total number of parameters to be estimated along the classification process is close to the total number of available samples, leaving us with fewer degrees of freedom. In the face of overfitting, whenever the least discriminating feature is discarded the cut point generalizes better than the line.

Regularization. The classical variance-bias tradeoff [24]–[26] and the more recent "double descent" literature [14], [15], [27]–[29] suggest that as the number of parameters in the model increases the error probability peaks around a point where model capacity roughly equals the number of samples. An isotropic case $\rho = 0$, $\delta = 1$ is discussed in [27]. In Table I we consider a non-isotropic case, and show that reducing the number of features has the effect of regularization, naturally leading to better generalization capability, which is critical for small sample sizes as overfitting must be avoided.

Linear regression. We leverage the error probabilities observed for $8 \leq n \leq 1024$ to obtain, through least squares regression, the following approximation,

$$E_1(n) = \alpha_1 + \beta_1/n, \quad E_2(n;\delta) = \alpha_2(\delta) + \beta_2(\delta)/n. \quad (6)$$

Whereas $L_d(\cdot)$ refers to the theoretical Bayes error and $E_d(n;\delta)$ denotes Monte Carlo estimates of the SVM test error, the regression model in (6) is an empirical approximation of $E_d(n;\delta)$. This choice is consistent with learning-theoretic results indicating that, for linear classifiers, the excess loss decays as $O(1/n)$ in the realizable case and as $O(1/\sqrt{n})$ in the agnostic case [30]–[34].

Table I reports the values of the intercepts and slopes for the single-feature model and for various values of δ defining the two-feature model. As expected, the intercepts agree up to two decimal digits with the asymptotic values reported in the line corresponding to $n = 1,024$ in Table I. In addition, intercepts and slopes increase with δ. Given these values, we readily obtain a closed-form expression to approximate the optimal threshold. It follows from (6) that $n^\star(\delta) = (\beta_2(\delta) - \beta_1)/(\alpha_1 - \alpha_2(\delta))$. Table I shows the values of $n^\star$ obtained from the above model, indicating its close agreement with SVM simulations.

B. When can samples compensate for features?

Figure 1, obtained from (6), shows fitted error curves as functions of the sample size n. The blue, green, and red curves correspond to two-feature classifiers with $\delta = 5$, 3, and 2, respectively, while the black curve corresponds to the one-feature case. For small n, the one-feature curve lies below the two-feature curves. At intermediate values of n, horizontal separations between curves (illustrated by Δn_1 and Δn_2) indicate that increasing the number of samples or increasing the number of features can yield comparable reductions in error. For larger n, the curves approach their respective asymptotic error levels.

Figure 1 provides a concrete illustration of the sample–feature tradeoff for $\delta = 2$. First, the red curve (two features, $\delta = 2$) intersects the black curve (one feature) at approximately $n = 12$, in agreement with Table I; thus, for $n < 12$ the one-feature classifier has lower error, whereas for $n \geq 12$ the two-feature classifier becomes preferable.

Second, the horizontal markers Δn_1 and Δn_2 highlight a local tradeoff between samples and features at a fixed error level. Around $n \approx 12$, increasing the sample size by only $\Delta n_2 = 2$ when using two features yields a reduction of about 3.25% in error probability. Achieving a comparable reduction with a single feature requires a substantially larger increase in sample size, approximately $\Delta n_1 = 10$.

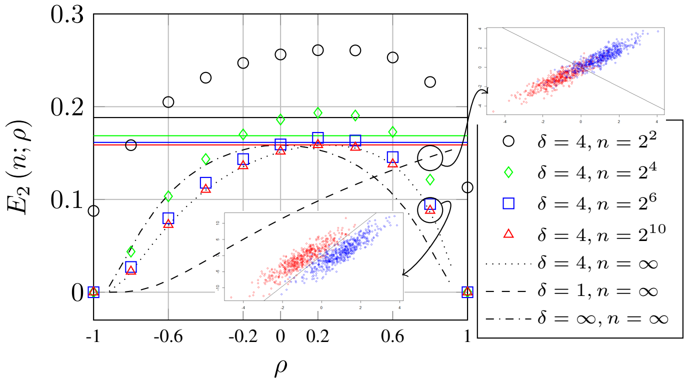

Fig. 2. Two-feature error probability $E_2(n; \rho)$ versus correlation ρ. Markers show SVM errors for $n = 4, 16, 64, 1024$ and $\delta = 4$. Dotted, dashed, and dash-dotted curves show Bayes-error predictions for $n = \infty$ with $\delta = 4$, $\delta = 1$, and $\delta \to \infty$, respectively (cf. (4)). The error curves are concave in ρ and peak near $\rho = 1/\delta$. The horizontal lines show the one-feature asymptotic error for $n = 4$ (top), $16, 64, 1024$ (bottom). Inset scatter plot illustrate representative joint distributions for $\delta = 4$ (below) and $\delta = 1$ (top).

Finally, the dashed horizontal lines indicate the asymptotic error levels. The black dashed line at $E = 0.159$ corresponds to the limiting error of the one-feature classifier, while the red dashed line at $E = 0.132$ corresponds to the limiting error of the two-feature classifier with $\delta = 2$. Therefore, any target error below 0.159 necessarily requires the use of two features, and errors approaching 0.132 can only be achieved asymptotically with sufficiently many samples.

C. Beyond independence: how does ρ impact $n^\star$?

We consider correlated features. For the covariance matrix in (2), we find that the crossover sample size $n^\star$, defined as the smallest n such that $E_2(n; \rho) < E_1(n)$, first increases and then decreases as ρ ranges from -1 to $+1$, as detailed next.

Figure 2 shows the two-feature classification error $E_2(n; \rho)$ as the correlation coefficient ρ varies, for $\delta = 4$. For finite sample sizes, each marker type corresponds to a different value of n ($n = 4, 16, 64, 1024$), and these points converge toward the corresponding asymptotic limits as n increases. The black curves give Bayes-error for $n = \infty$ from (4) for $\delta \in \{4, 1, \infty\}$, while the horizontal lines indicate the one-feature errors.

Across all cases, the error is a concave function of ρ, with a maximum near $\rho = 1/\delta$, where the Bayes error attains its largest value. This region corresponds to a regime in which the second feature provides little additional discriminative power, so improvements from using two features become visible only at large n. In contrast, for ρ near ± 1, the error is substantially lower, reflecting strong linear dependence between features and improved separability. Indeed, for $\rho = 1$ and $\rho = -1$ having the second feature is always helpful. As $\rho \to 1/\delta$, the second feature is beneficial for $n \geq 1024$.

Intuition. When $\rho = 1/\delta$ the best separator of the two classes corresponds to a vertical line, $X_1 = 0$, i.e., feature X_2 is useless. In general, the best separator is $X_2 = \tilde{b} X_1$. As ρ approaches $1/\delta$ from below (resp., above), $\tilde{b}$ tends to $-\infty$ (resp., $+\infty$) and the contribution of X_2 to the best separator decreases, causing a corresponding increase in classification error (cf. (5)). When $\rho = \pm 1$, in contrast, the best slope is $\tilde{b} = \pm \delta$, i.e., X_2 contributes to the best separator given by $X_1 = \pm X_2 / \delta$ (see Section III).

TABLE II

ERROR PROBABILITIES (MULTIPLIED BY 100). DIFFERENCES Δ_ρ HIGHLIGHT WHERE 3 FEATURES OUTPERFORM 2 FEATURES.

		3 features					
		$\rho_{13}{=}\rho_{23}{=}0.0$		$\rho_{13}{=}\rho_{23}{=}0.3$		$\rho_{13}{=}\rho_{23}{=}0.4$	
n	2 feat	$E_3(n)$	$\Delta_0(n)$	$E_3(n)$	$\Delta_{0.3}(n)$	$E_3(n)$	$\Delta_{0.4}(n)$
2	14.36	11.09	-3.265	13.48	-0.876	14.20	-0.162
4	12.94	9.55	-3.393	12.62	-0.322	13.36	+0.423
8	10.84	7.97	-2.872	11.07	+0.230	11.69	+0.850
16	9.09	6.37	-2.718	9.24	+0.151	9.82	+0.727
32	8.05	5.25	-2.794	7.99	-0.061	8.52	+0.477
64	7.47	4.51	-2.960	7.22	-0.257	7.74	+0.264
128	7.16	4.11	-3.049	6.75	-0.406	7.26	+0.106
256	6.98	3.88	-3.103	6.48	-0.497	6.98	+0.004
512	6.89	3.77	-3.125	6.36	-0.530	6.86	-0.030
1024	6.85	3.70	-3.147	6.29	-0.558	6.79	-0.061

Figure 2 also shows that when $\rho \neq 0$ an increase in δ may be beneficial. Indeed, for $\delta_1 < \delta_2$ and $\rho_0 = (\delta_1 + \delta_2)/(2\delta_1\delta_2)$ we have that $E_2(\infty; \delta_1, \rho) \lesseqgtr E_2(\infty; \delta_2, \rho)$ if $\rho \lesseqgtr \rho_0$. As illustrated in the scatter plots in Figure 2, if $\rho > 0$ an increase in the dispersion of the points may simplify their separation. In particular, for $\rho = 0.8$, the scatter plots show that the two classes are more clearly separated when $\delta = 4$ than when $\delta = 1$, consistent with the lower error for $\delta = 4$ at $\rho = 0.8$.

Different covariance matrices. When each class corresponds to a different covariance matrix, classification of samples may be exclusively based on covariances, rendering both features necessary. This is the case when features are functionally dependent on each other, e.g., $X_1 = -X_2$ and $X_1 = X_2$ for classes 1 and 2, respectively. In that case, $n^\star = 0$ and samples cannot compensate for features.

Multi-dimensional extension. We now extend the analysis to three features. Let $\sigma_1 = \sigma_2 = \sigma_3 = 1$, $\rho_{12} = -0.1$, $\rho_{13} = \rho_{23} = 0.3$, and class means $(+1, +1, +1)$ and $(-1, -1, -1)$. In this regime, the usefulness of adding the third feature X_3 cannot be characterized by a single sample-size threshold. Indeed, Table II shows that the ordering between the two- and three-feature errors changes twice as n increases: $E_3(n) < E_2(n)$ at $n = 4$, but $E_3(n) > E_2(n)$ at $n = 8$ and $n = 16$, and then $E_3(n) < E_2(n)$ again from $n = 32$ onward. Consequently, the corresponding error curves must cross once for some $n \in [4, 8]$ and a second time for some $n \in [16, 32]$. When $\rho_{13} = \rho_{23} = 0.4$, Table II reveals a similar non-monotone comparison. These multiple crossings stand in stark contrast to the two-dimensional case, where we observed at most one crossing. A deeper analysis of this phenomenon is left for future work.

V. Discussion and Conclusion

Understanding how the classification error probability decays with the sample size is central in statistical learning theory [35]–[43]. In this paper, we study this decay in a tractable setting, characterizing the *joint* effect of variability and dependence on the classification error probability [9], [10].

Our work is part of a research agenda on distributed learning under constraints [24]–[29]. Here we considered classification tasks, and in [44] we considered estimation, leaving additional tasks, e.g., change point detection [45], as subject for future work. In addition, we assume i.i.d. sampling; studying non-uniform policies (e.g., importance sampling) under sensing/communication constraints is also left for future work [46].

REFERENCES

[1] S. Zhang, X. Kang, P. Duan, B. Sun, and S. Li, “Polygon structure-guided hyperspectral image classification with single sample for strong geometric characteristics scenes,” *IEEE Transactions on Geoscience and Remote Sensing*, 2021.

[2] F. He, K. Lv, J. Yang, and X. Huang, “One-Shot Distributed Algorithm for PCA With RBF Kernels,” *IEEE Signal Processing Letters*, vol. 28, pp. 1465–1469, 2021.

[3] B. Hanczar and E. R. Dougherty, “The reliability of estimated confidence intervals for classification error rates when only a single sample is available,” *Pattern Recognition*, vol. 46, no. 3, pp. 1067–1077, 2013.

[4] E. G. Miller, N. E. Matsakis, and P. A. Viola, “Learning from one example through shared densities on transforms,” in *Proceedings IEEE Conference on Computer Vision and Pattern Recognition. CVPR 2000 (Cat. No. PR00662)*, vol. 1. IEEE, 2000, pp. 464–471.

[5] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, “Generalizing from a few examples: A survey on few-shot learning,” *ACM Computing Surveys (CSUR)*, vol. 53, no. 3, pp. 1–34, 2020.

[6] H. Noh, R. Rajagopal, and A. Kiremidjian, “Sequential structural damage diagnosis algorithm using a change point detection method,” *Journal of Sound and Vibration*, vol. 332, no. 24, pp. 6419–6433, 2013.

[7] P.-N. Chen and A. Papamarcou, “Likelihood ratio partitions for distributed signal detection in correlated Gaussian noise,” in *Proceedings of 1995 IEEE International Symposium on Information Theory*. IEEE, 1995, p. 118.

[8] P. Willett, P. F. Swaszek, and R. S. Blum, “The good, bad and ugly: distributed detection of a known signal in dependent gaussian noise,” *IEEE Transactions on signal processing*, vol. 48, no. 12, pp. 3266–3279, 2000.

[9] G. Hughes, “On the mean accuracy of statistical pattern recognizers,” *IEEE Transactions on Information Theory*, vol. 14, no. 1, pp. 55–63, 1968.

[10] J. Hua, Z. Xiong, J. Lowey, E. Suh, and E. R. Dougherty, “Optimal number of features as a function of sample size for various classification rules,” *Bioinformatics*, vol. 21, no. 8, pp. 1509–1515, 2005.

[11] P. R. Azevedo, “SLACGS: Simulator for Loss Analysis of Classifiers using Gaussian Samples,” 2025, version 1.0. GitHub repository. [Online]. Available: https://github.com/paulorenatoaz/slacgs

[12] L. Devroye, L. Györfi, and G. Lugosi, *A probabilistic theory of pattern recognition*. Springer Science & Business Media, 2013, vol. 31.

[13] A. Faragó and G. Lugosi, “Strong universal consistency of neural network classifiers,” *IEEE Transactions on Information Theory*, vol. 39, no. 4, pp. 1146–1151, 1993.

[14] P. Nakkiran, P. Venkat, S. M. Kakade, and T. Ma, “Optimal regularization can mitigate double descent,” in *International Conference on Learning Representations*, 2020.

[15] S. d’Ascoli, L. Sagun, and G. Biroli, “Triple descent and the two kinds of overfitting: where & why do they appear?” in *NeurIPS*, 2020.

[16] G. Blanchard, O. Bousquet, and P. Massart, “Statistical performance of support vector machines,” *The Annals of Statistics*, vol. 36, no. 2, pp. 489–531, 2008.

[17] L. Bottou and C.-J. Lin, “Support vector machine solvers,” *Large scale kernel machines*, vol. 3, no. 1, pp. 301–320, 2007.

[18] N. List and H. U. Simon, “SVM-optimization and steepest-descent line search,” in *Proceedings of the 22nd Annual Conference on Computational Learning Theory*. Citeseer, 2009.

[19] S. Ghosh and K. N. Chaudhury, “Fast and high-quality bilateral filtering using Gauss–Chebyshev approximation,” in *Proc. International Conference on Signal Processing and Communications (SPCOM)*, Bangalore, India.

[20] C.-J. Hsieh, K.-W. Chang, C.-J. Lin, S. S. Keerthi, and S. Sundararajan, “A dual coordinate descent method for large-scale linear SVM,” in *Proceedings of the 25th international conference on Machine learning*, 2008, pp. 408–415.

[21] T. Glasmachers, “Universal consistency of multi-class support vector classification,” *Advances in Neural Information Processing Systems*, vol. 23, pp. 739–747, 2010.

[22] R. Vert, J.-P. Vert, and B. Schölkopf, “Consistency and convergence rates of one-class SVMs and related algorithms.” *Journal of Machine Learning Research*, vol. 7, no. 5, 2006.

[23] “R documentation,” 2021, https://www.rdocumentation.org/packages/e1071.

[24] R. M. Hogarth and N. Karelaia, “Ignoring information in binary choice with continuous variables: When is less “more”?” *Journal of Mathematical Psychology*, vol. 49, no. 2, pp. 115–124, 2005.

[25] D. G. Goldstein and G. Gigerenzer, “Fast and frugal forecasting,” *International journal of forecasting*, vol. 25, no. 4, pp. 760–772, 2009.

[26] P. Domingos, “A unified bias-variance decomposition,” in *Proceedings of 17th International Conference on Machine Learning*, 2000, pp. 231–238.

[27] Z. Deng, A. Kammoun, and C. Thrampoulidis, “A model of double descent for high-dimensional binary linear classification,” *Information and Inference: A Journal of the IMA*, 2021.

[28] M. Loog, T. Viering, and A. Mey, “Minimizers of the empirical risk and risk monotonicity,” *Advances in Neural Information Processing Systems*, vol. 32, pp. 7478–7487, 2019.

[29] M. Belkin, D. Hsu, S. Ma, and S. Mandal, “Reconciling modern machine-learning practice and the classical bias–variance trade-off,” *Proceedings of the National Academy of Sciences*, vol. 116, no. 32, pp. 15 849–15 854, 2019.

[30] C. Cortes, L. D. Jackel, S. A. Solla, V. Vapnik, and J. S. Denker, “Learning curves: Asymptotic values and rate of convergence,” in *Advances in Neural Information Processing Systems*, 1994, pp. 327–334.

[31] H. S. Seung, H. Sompolinsky, and N. Tishby, “Statistical mechanics of learning from examples,” *Physical review A*, vol. 45, no. 8, p. 6056, 1992.

[32] N. Srebro and S. Ben-David, “Learning bounds for support vector machines with learned kernels,” in *International Conference on Computational Learning Theory*. Springer, 2006, pp. 169–183.

[33] R. F. de Mello, C. Manapragada, and A. Bifet, “Measuring the shattering coefficient of decision tree models,” *Expert Systems with Applications*, vol. 137, pp. 443–452, 2019.

[34] S. Shalev-Shwartz and S. Ben-David, *Understanding machine learning: From theory to algorithms*. Cambridge university press, 2014.

[35] C. Giraud, “Introduction to high-dimensional statistics,” *Monographs on Statistics and Applied Probability*, vol. 139, p. 139, 2015.

[36] Y.-Z. J. Chen, D. S. Menasché, and D. Towsley, “To collaborate or not in distributed statistical estimation with resource constraints?” in *2021 55th Annual Conference on Information Sciences and Systems (CISS)*. IEEE, 2021, pp. 1–6.

[37] Y. S. Abu-Mostafa, M. Magdon-Ismail, and H.-T. Lin, *Learning from data*. AMLBook New York, NY, USA:, 2012, vol. 4.

[38] J. Friedman, T. Hastie, R. Tibshirani *et al.*, *The elements of statistical learning*. Springer series in statistics New York, 2001, vol. 1, no. 10.

[39] R. F. de Mello, “On the shattering coefficient of supervised learning algorithms,” *arXiv preprint arXiv:1911.05461*, 2019.

[40] R. F. Mello and M. A. Ponti, *Machine learning: a practical approach on the statistical learning theory*. Springer, 2018.

[41] T. Cover and J. Thomas, *Elements of information theory*. John Wiley & Sons, 1999.

[42] J. Gareth, D. Witten, T. Hastie, and R. Tibshirani, *An introduction to statistical learning: with applications in R*. Spinger, 2013.

[43] V. Berisha, A. Wisler, A. O. Hero, and A. Spanias, “Empirically estimable classification bounds based on a nonparametric divergence measure,” *IEEE Transactions on Signal Processing*, vol. 64, no. 3, pp. 580–591, 2015.

[44] Y.-Z. J. Chen, D. S. Menasché, and D. Towsley, “On collaboration in distributed parameter estimation with resource constraints,” *IEEE Transactions on Network and Service Management*, 2024.

[45] A. R. Ramtin, P. Nain, and D. Towsley, “Quickest change detection in continuous-time in presence of a covert adversary,” *IEEE Signal Processing Letters*, 2025.

[46] S. Ghosh and K. N. Chaudhury, “Color bilateral filtering using stratified Fourier sampling,” in *Proc. IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, California, USA, 2018, pp. 26–30.